

Empowering Compute, Network, Storage & AI

DDR5 Memory Validation Report

Server 172.16.13.19 - Two Campaigns (256 GB + 128 GB)

Server / Host	172.16.13.19 (172.16.13.19)
Platform	Tyrone MDI300 / 2x Intel Xeon 6730P (64C/128T, Granite Rapids)
Validation Tier	Qualification
Campaign Window	28 May 2026 (A 08:25-09:50, B 10:13-10:58 UTC)
Prepared For	Netweb Technologies India Limited
Date	2026-05-28

Overall Result: PASS (MEMORY), ACTION (PLATFORM - CPU 500 MHz)

Executive Summary

Two memory configurations were validated back-to-back on the same server 172.16.13.19 (Tyrone MDI300, dual Intel Xeon 6730P Granite Rapids): Campaign A - 8x Samsung 256 GB DDR5-6400 (2 TB); Campaign B - 8x Samsung 128 GB DDR5-6400 (1 TB) after a physical swap.

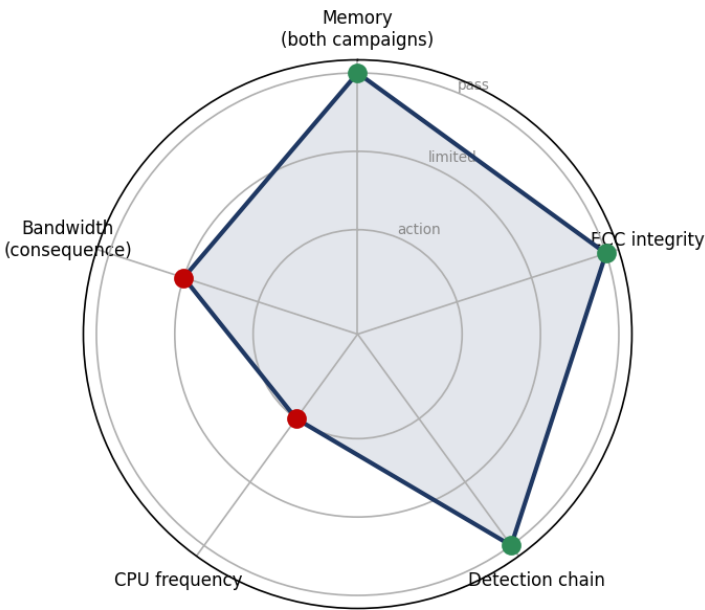
The memory passes in both campaigns: all 8 modules detected by BMC, BIOS and kernel, trained at full rated 6400 MT/s, with zero ECC errors across a 30-minute pattern-verify pass and a 15-minute >=95%-RAM-resident burn each.

Testing also identified a platform firmware fault: every CPU core is locked at 500 MHz in BOTH campaigns. The lock survived the reboot and DIMM swap - confirming it is independent of the memory. It must be fixed before this server is benchmarked or deployed; bandwidth figures here are throttled ~10x and are not representative.

Validation Scorecard

Subsystem	Result	Detail	Verdict
Memory (both campaigns)	0 ECC, 8/8 DIMMs, full 6400 MT/s	30-min verify + 15-min >95% burn, 0 failures	PASS
ECC integrity	0 CE / 0 UE on all 16 controllers	0 MCE; mlock succeeded on all 64 workers	PASS
Detection chain	BMC + BIOS + kernel agree: 8 DIMMs	Multi-bit ECC active; correct SPD/speed	PASS
CPU frequency	Locked at 500 MHz (all 64 cores)	vs 2500 base / 3800 turbo; firmware fault, not thermal/power	ACTION
Bandwidth (consequence)	~23 GB/s aggregate (throttled ~10x)	Not representative until the CPU lock is cleared	LIMITED

Subsystem health at a glance



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	MDI300
Hostname	172.16.13.19

CPU

Model	2x Intel Xeon 6730P (Granite Rapids)
Sockets	2
Cores Per Socket	32
Threads	128
Base Ghz	2.5
Boost Ghz	3.8
Numa Nodes	2
Isa	AVX-512, AMX (L3 576 MiB)

Memory

Total	2 TB (A) / 1 TB (B)
Type	DDR5 (Samsung)
Dimms Populated	8 of 32 (4-of-8 channels: P0/P1 A,B,E,F)
Configured Speed	6400 MT/s
Rated Speed	6400 MT/s
Speed Status	full rated speed (not clamped)

BIOS / Platform

Version	AMI ES418INW.M06
Date	2026-04-16

BMC / Power

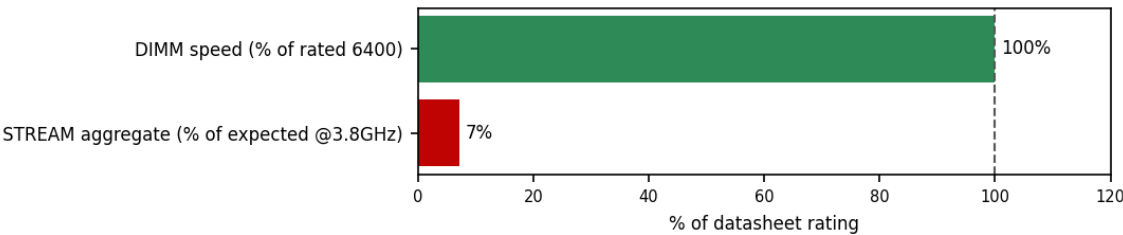
Firmware	1.11
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Kernel / OS

Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-111-generic

Specification Compliance

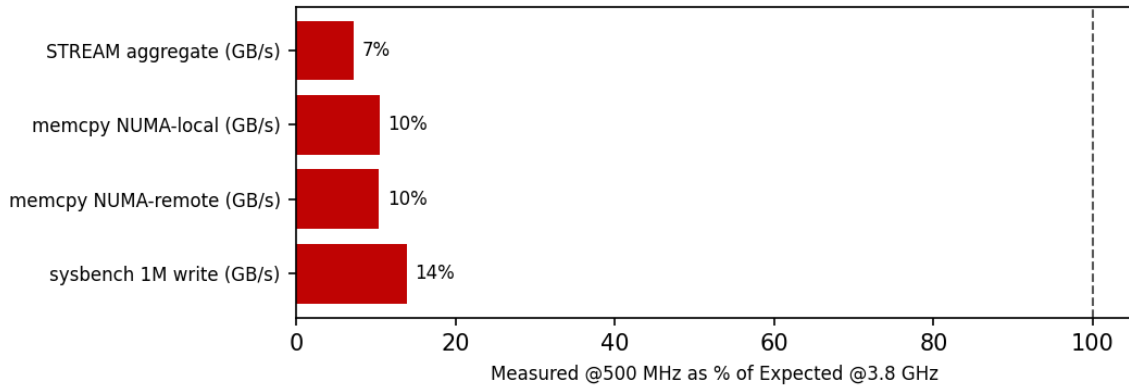
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
DIMM speed (% of rated 6400)	6,400	6,400	MT/s	100%
STREAM aggregate (% of expected @3.8GHz)	23	320	GB/s	7%

Cross-Server Comparison

Measured bandwidth vs expected at rated frequency.



Metric	Expected @3.8 GHz	Measured @500 MHz	Delta
STREAM aggregate (GB/s)	320	23	-93%
memcpy NUMA-local (GB/s)	22.00	2.31	-90%
memcpy NUMA-remote (GB/s)	13.50	1.40	-90%
sysbench 1M write (GB/s)	17.50	2.42	-86%

Bandwidth is throttled ~10x by the 500 MHz CPU lock and is NOT representative - measured is only ~7-14% of expected. Both campaigns produce nearly identical throttled numbers, further proof the bottleneck is the CPU lock, not the DIMMs. Re-measure after the lock is cleared.

Component Health

BMC SEL (before → after): BMC SEL: 0 memory-related events. IPMI sees exactly 8 populated DIMM thermal sensors (28-32 C), all 'ok'.

Kernel log (dmesg) diff: 0 MCE; EDAC enumerated all 16 controllers, CE=0 / UE=0 in both campaigns. turbostat shows all 64 cores pinned at 500 MHz despite the OS requesting 3.8 GHz (HWP/PERF_CTL ratio 0x26); no thermal or power limit active (29-35 C, 88-139 W vs 250 W cap, CORE_PERF_LIMIT_REASONS=0). dmesg: intel_uncore_frequency_tpmi 'Unsupported minor version' (kernel 6.8 PM support incomplete for Granite Rapids).

Findings & Recommendations

Findings

- Both DIMM sets (8x 256 GB + 8x 128 GB Samsung DDR5-6400) are healthy and fully validated: detected by BMC/BIOS/kernel, trained at full rated 6400 MT/s, zero ECC errors, zero MCE, mlock on all 64 workers.
- PLATFORM FAULT: every CPU core is locked at 500 MHz (one-fifth of the 2500 MHz base) in both campaigns. The lock persisted across reboot and DIMM swap - it is a firmware-level condition independent of the memory.
- The OS requests 3.8 GHz through every mechanism and the silicon reports it is capable, with no thermal or power limit active - pointing to a BIOS power-profile / config-TDP setting or early-Granite-Rapids pcode/microcode immaturity.
- Bandwidth is throttled ~10x by the CPU lock (~23 GB/s aggregate vs ~300+ expected) - recorded for completeness only, not representative of the platform.

Recommendations

- Memory: cleared for use - both the 256 GB and 128 GB Samsung DDR5-6400 sets pass all correctness and detection checks. No DIMM action required.
- Resolve the 500 MHz CPU lock before benchmarking/deployment (in order of likelihood): (a) BIOS power profile -> set Performance / max TDP; (b) update BIOS + Intel microcode (Granite Rapids pcode); (c) check VR / power-delivery telemetry; (d) try a newer kernel (6.11+/HWE) for full PM support.
- Re-run the bandwidth suite once cores reach rated frequency - expect ~300+ GB/s aggregate STREAM for this 4-of-8-channel DDR5-6400 layout.

Conclusion

Area	Outcome	Verdict
8x 256 GB + 8x 128 GB DDR5-6400	Memory validated: 0 ECC, full 6400 MT/s, >95% burn (both campaigns)	PASS
CPU frequency lock (500 MHz)	Firmware fault in both campaigns; fix BIOS power profile / microcode before deployment	ACTION
Performance benchmarking	Blocked until the CPU lock is cleared; bandwidth not representative	LIMITED

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Stress Ng	stress-ng --vm 64 --verify (30 min)
Memtester	memtester (64 workers, 15 min)
Bandwidth	mbw memcpy; stress-ng --stream; sysbench memory
Cpu Freq	turbostat; rdmsr HWP_REQUEST/PERF_CTL/HWP_CAPABILITIES
Edac	16 memory controllers, multi-bit ECC
Bios	AMI ES418INW.M06 (2026-04-16)
Bmc Firmware	1.11
Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-111-generic
Campaign Window	28 May 2026 (A 08:25-09:50, B 10:13-10:58 UTC)
Note	Reconstructed from the validated source report into the CoreBench results.json contract.